L7

CLASS X - SCIENCE

# LIFE PROCESSES

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# LET'S TEST YOUR MEMORY

Q. Answer the following questions based on the diagram given below:

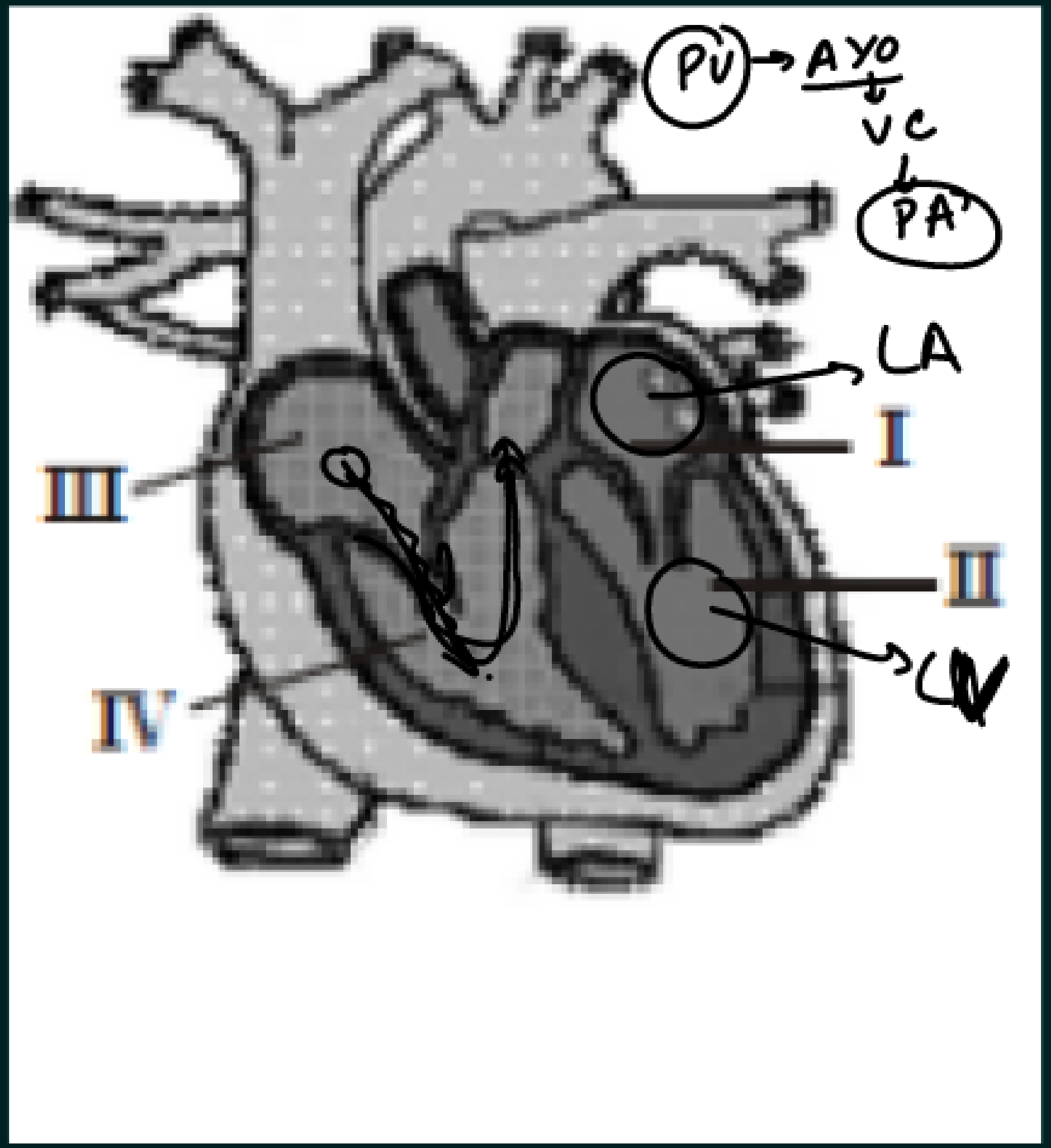
(i) Label the four chambers of the heart in the diagram.

(ii) What is the function of the valves shown in the heart diagram?

(iii) Which side of the heart pumps oxygenated blood to the body, and which side pumps deoxygenated blood to the lungs?

(iv) Describe the pathway of blood through the heart starting from the right atrium.

(v) Why does the left ventricle have a thicker muscular wall compared to the right ventricle?



# LET'S TEST YOUR MEMORY

**Q: Why do more chambers in the heart matter in evolution?**

More heart chambers help in better separation of oxygenated and deoxygenated blood. This increases oxygen supply to cells, improves respiration, produces more ATP (energy), and allows warm-blooded animals to maintain body temperature. It makes circulation more efficient, helping organisms survive better.

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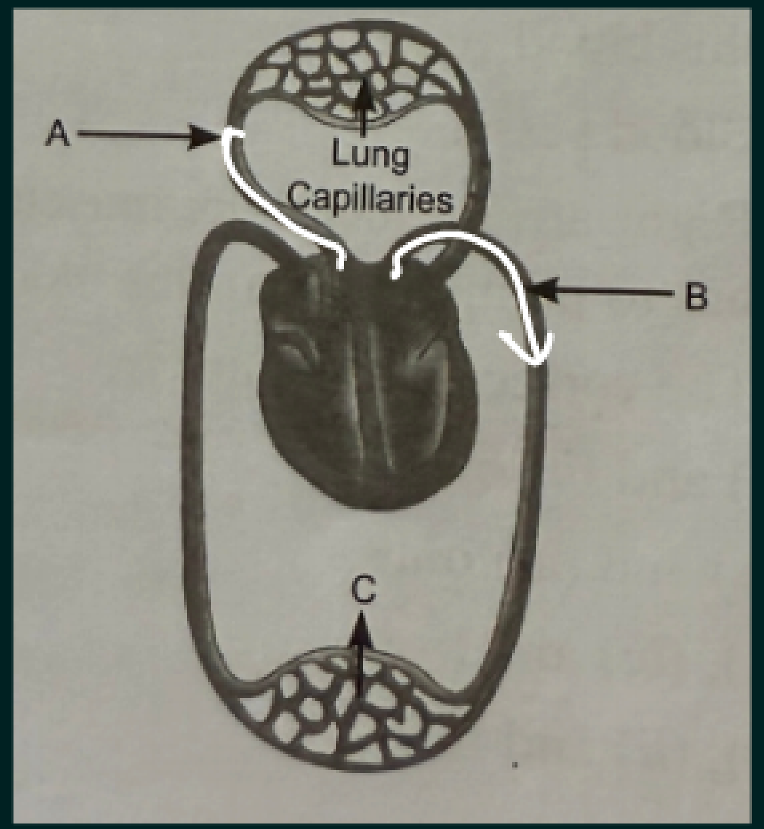
# LET'S TEST YOUR MEMORY

Q. Study the diagram given below showing the schematic representation of transport and exchange of gases in the human heart.

(a) Name the parts labelled A, B and C in the diagram and mention one function of each.

(b) Explain the significance of having four chambers in the human heart.?




(a) A – Pulmonary artery : Carries deoxygenated blood from the heart to the lungs.

B – Aorta : Carries oxygenated blood from the heart to different parts of the body.

C - Capillaries: Help in the exchange of materials (oxygen, carbon dioxide, nutrients, and wastes) between blood and surrounding tissues.

(b) The human heart has four chambers—two atria and two ventricles—which play an important role in efficient circulation of blood.

- The right side of the heart carries deoxygenated blood to the lungs, while the left side carries oxygenated blood to the body.
- The separation of the right and left sides prevents mixing of oxygenated and deoxygenated blood.
- This ensures that tissues receive blood rich in oxygen, which is necessary for releasing more energy.
- Efficient oxygen supply helps in maintaining a high metabolic rate, required for activities and maintaining constant body temperature.

Thus, a four-chambered heart makes circulation more efficient and effective in human beings.

# Lymph:

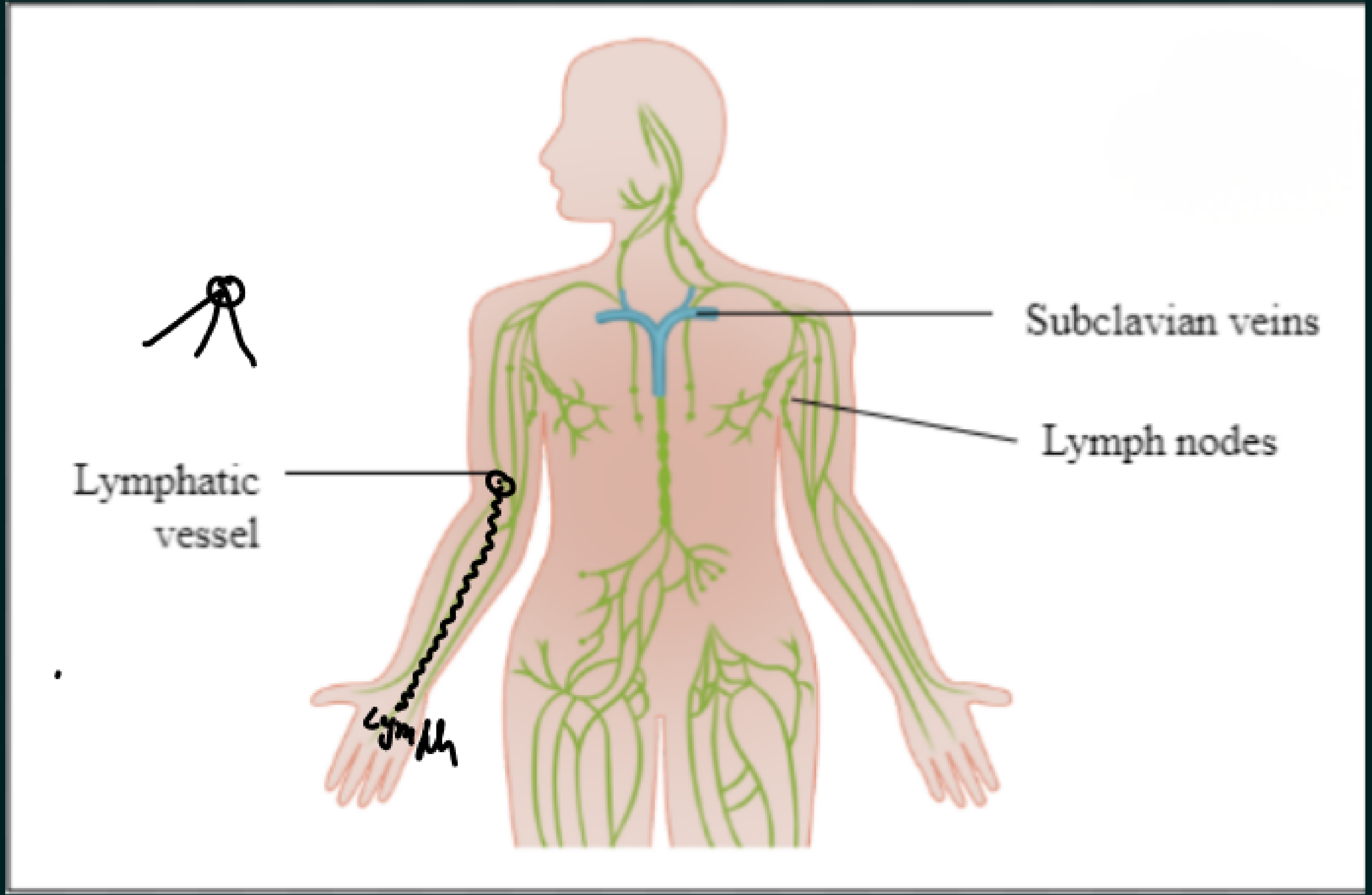
*Lymph is also called tissue fluid. It is a colourless fluid involved in transportation.* → White

**Formation of Lymph:**

- Some amount of plasma, proteins and blood cells escape through the pores present in the walls of capillaries.
- This fluid enters the intercellular spaces in tissues and forms lymph.

**Nature of Lymph:** Lymph is similar to blood plasma but is colourless and contains less protein.

1. *Lymph fluid*
2. *Lymph vessels*
3. *Lymph nodes*



## Flow of Lymph:

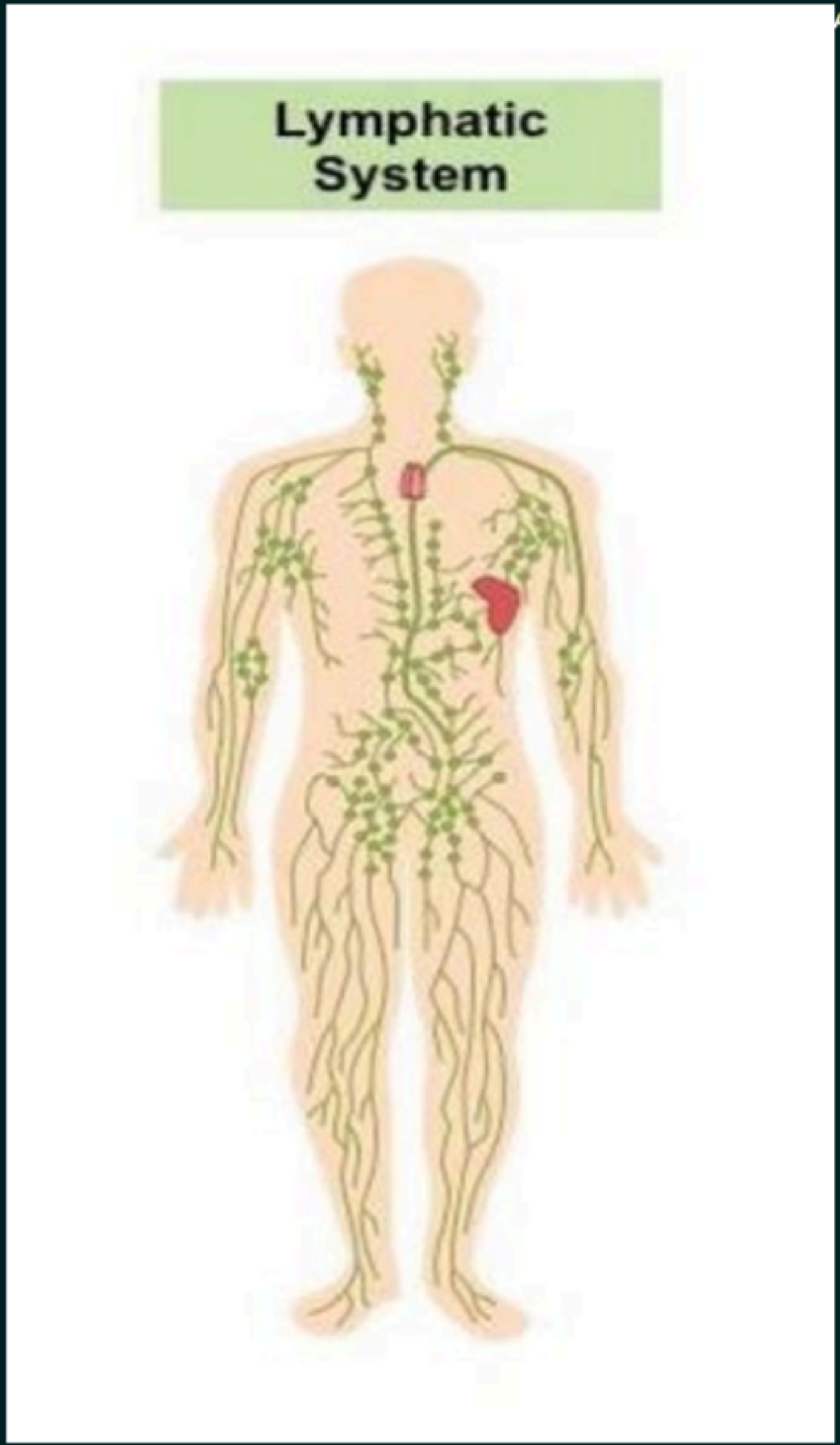
- Lymph drains from intercellular spaces into lymphatic capillaries.
- These lymphatic capillaries join to form large lymph vessels.
- The large lymph vessels finally open into larger veins.

## Functions of Lymph:

to Blood

1 - Lymph carries digested and absorbed fat from the intestine.
2 - It drains excess fluid from extracellular/intercellular spaces back into the blood.

3) Lymph has WBC → Immunity



| Feature | Blood | Lymph |
|---|---|---|
| **Colour** | Red due to the presence of haemoglobin in RBCs. | Colourless or pale yellow because it does not contain RBCs. |
| **Haemoglobin** | Present. | Absent. |
| **Main components** | Plasma, RBCs, WBCs and platelets. | Similar to plasma but has fewer proteins and mainly contains WBCs. |
| **Flow** | Flows inside blood vessels like arteries, veins and capillaries. | Flows in lymph vessels. |
| **Function** | Transports oxygen, carbon dioxide, nutrients, hormones and wastes between organs. | Carries digested fats from intestine and drains extra fluid from tissues back into blood. |
| **Clotting** | Platelets help in clotting of blood. | Clotting is very slow or less prominent because platelets are absent or very few. |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Transportation in
Plants

# Transportation in Plants

Transportation is the process by which plants move essential materials like oxygen, food, salts, carbon dioxide, and waste between different parts of their body to stay alive and function properly.

***Need of Transportation in Plants:***

- **Plants take in carbon dioxide and prepare food by photosynthesis in their leaves.**
- **But plants also need raw materials like nitrogen, phosphorus and other minerals.**
- **These raw materials are absorbed from the soil through the roots.**

# Why Diffusion is Not Enough

When the distance between two plant cells is small, water and nutrients can move easily through diffusion and osmosis.

- If the distance between roots and leaves is small, substances can move by diffusion.
- But in large plants and tall trees, the distance between roots and leaves is very large.
- So, diffusion alone is not sufficient to transport raw materials and food to all parts of the plant.
- Therefore, plants need a proper transportation system.

# Energy Needs in Plants

- Plants do not move from one place to another.
- Many tissues in plants have a large proportion of dead cells.
- So, plants have low energy needs.
- Therefore, plants can use relatively slow transport systems.

# Transport of Water and Minerals – by Xylem

Xylem transports water and minerals from roots to leaves.

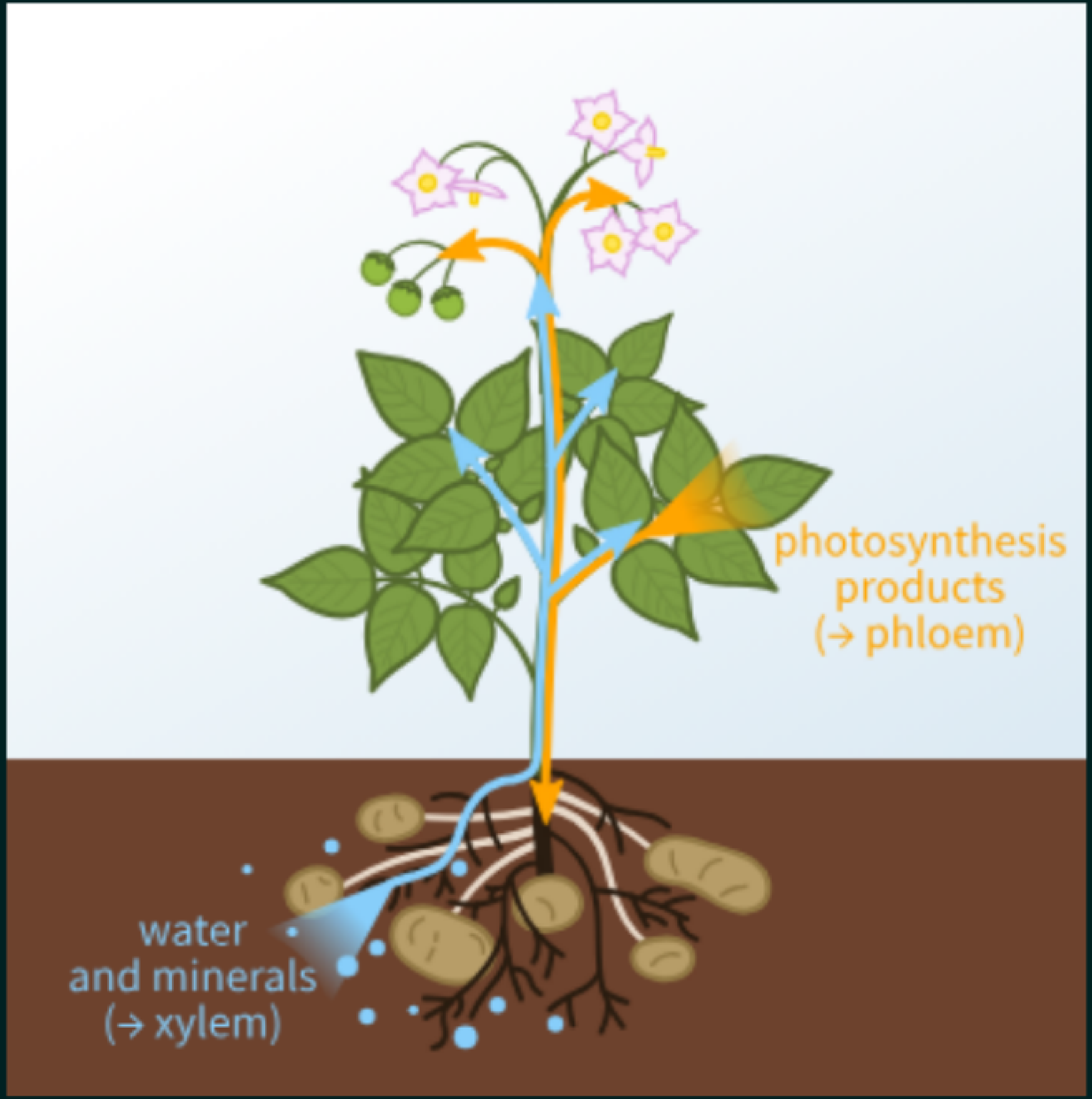
Water enters the roots from the soil and moves upward through xylem vessels.

This upward movement of water and minerals through xylem is called *ascent of sap*.

Ascent of sap is the upward movement of water and minerals from the roots to the upper parts of the plant through xylem.

This movement takes place mainly due to:

1. Root pressure
2. Transpiration pull — major force during the day.

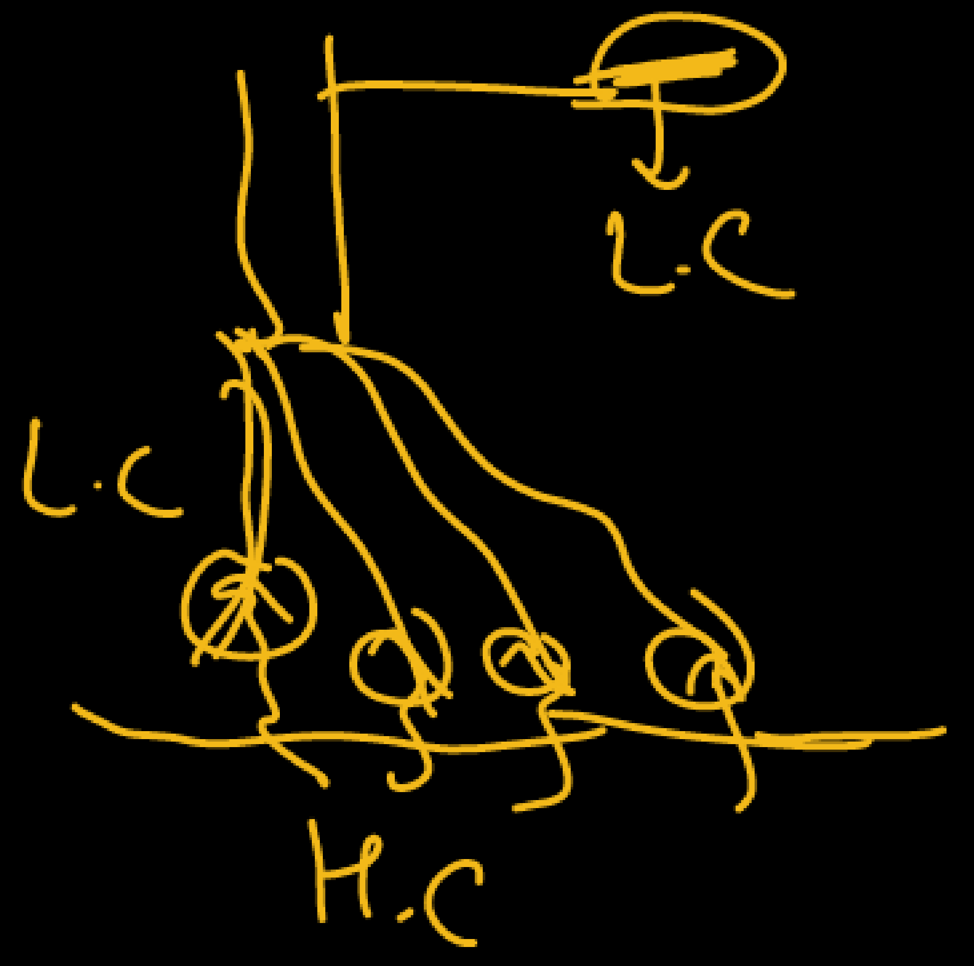


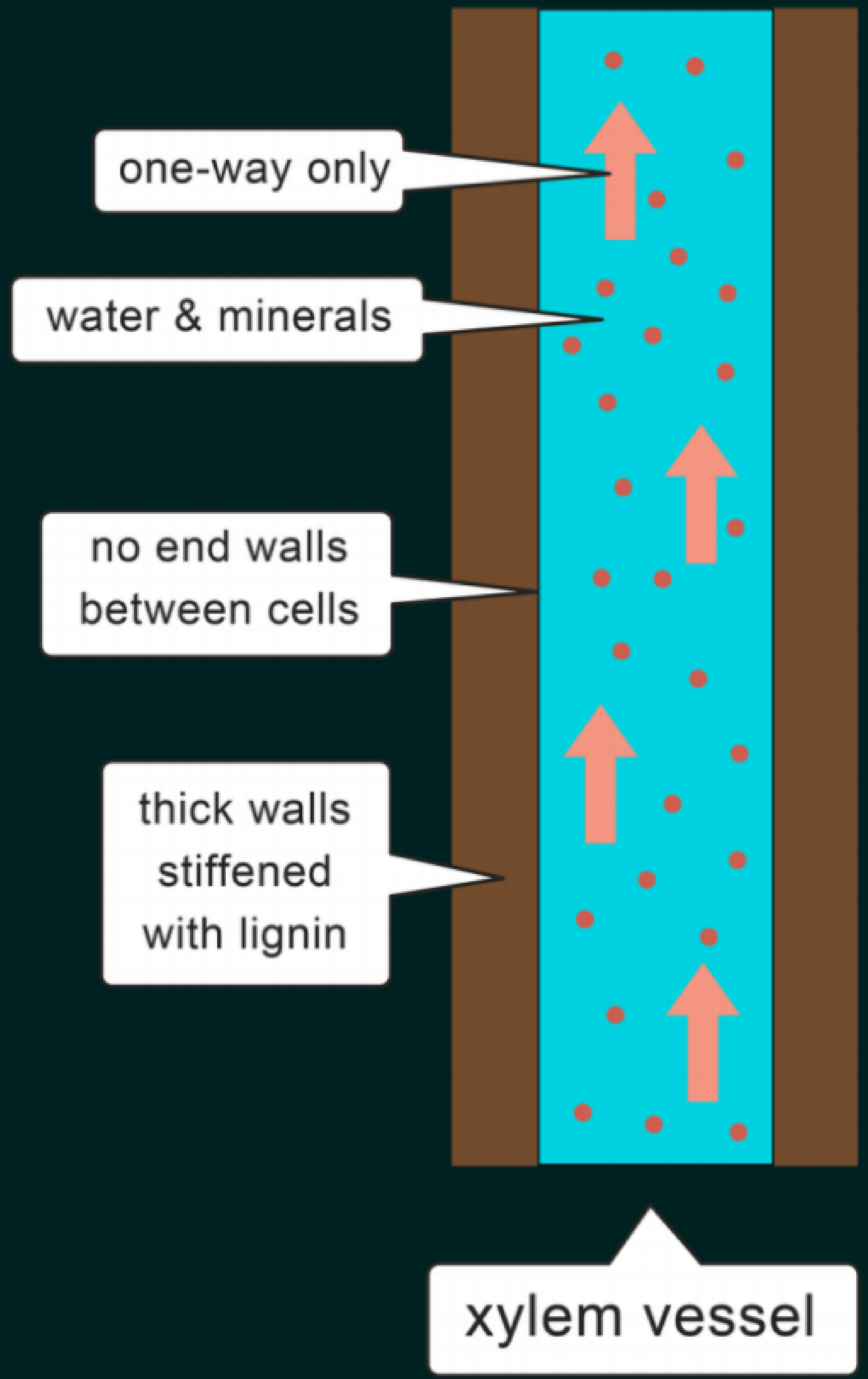
Root pressure

Transpiration

Day → R.P + T

Night → R.P

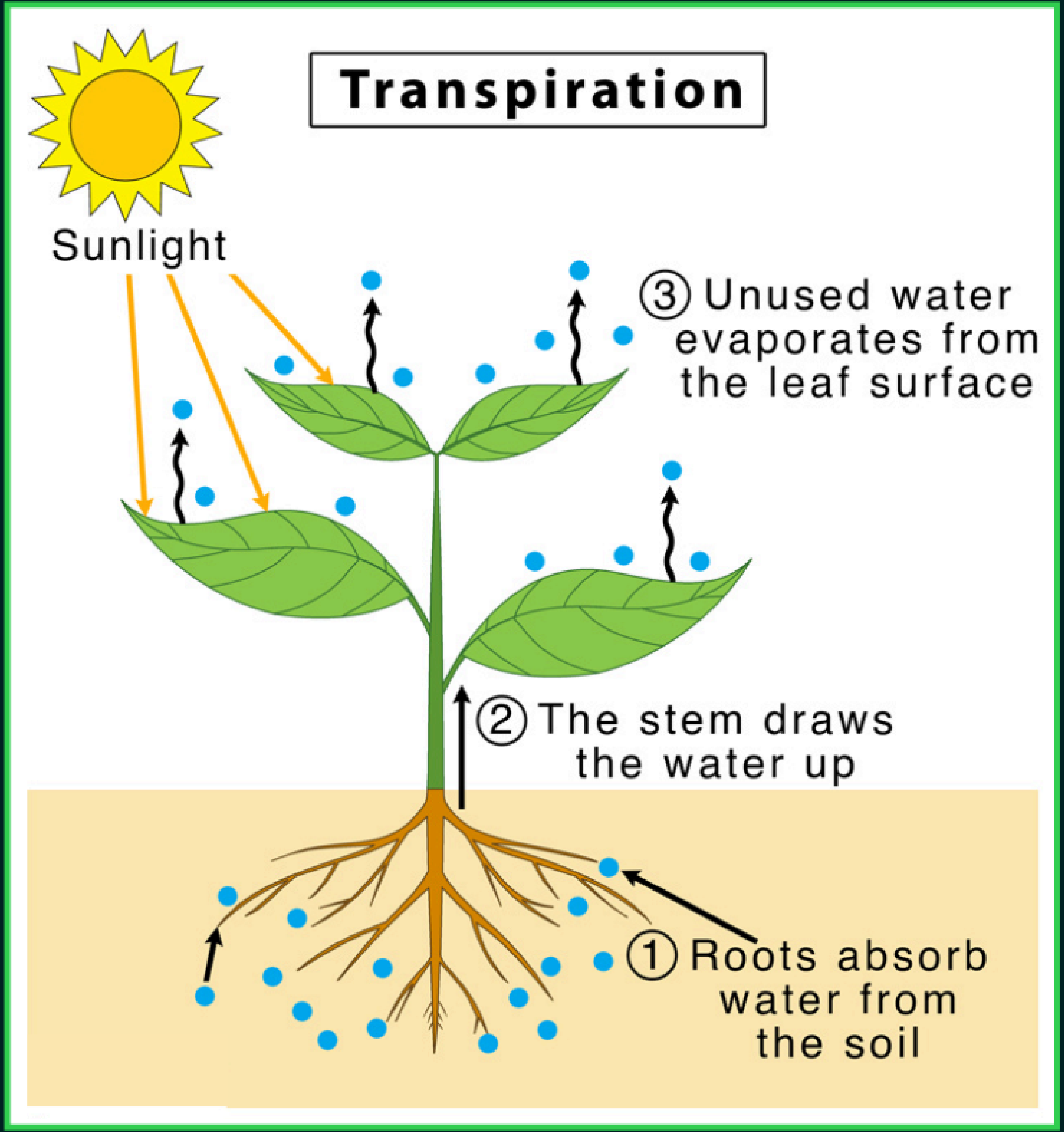





Xylem is a One-Way Highway.
It flows water and minerals only in the upward direction.

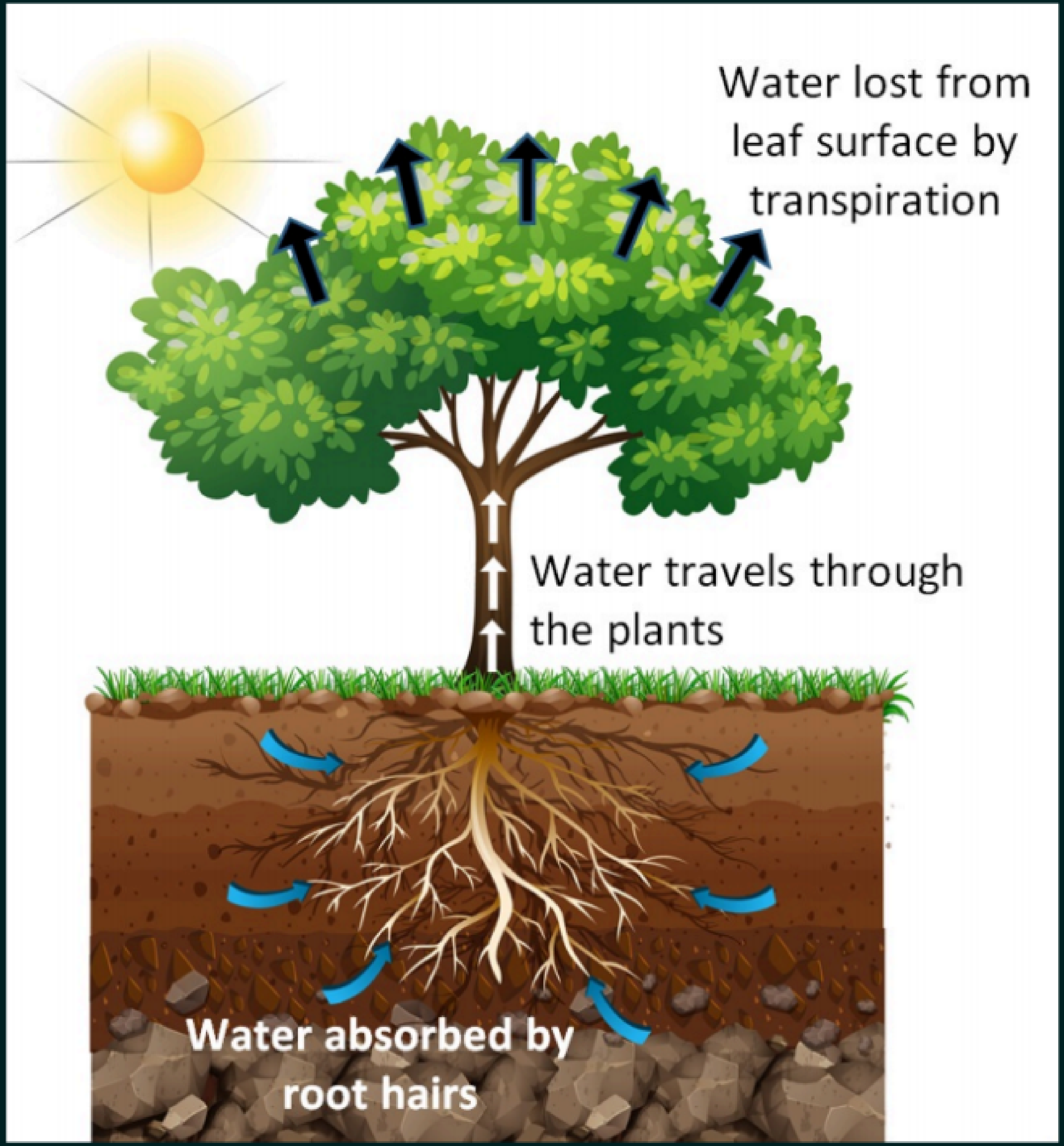
# Transpiration

Transpiration is the process by which plants lose water in the form of water vapour from the aerial parts, especially through tiny openings called stomata on the leaves.



# Transpiration / Suction Pull

- When water evaporates from the cells of leaves, it creates a suction pull.
- This suction pull draws water upward from the xylem cells of roots to the leaves.
- This upward movement of water is called **transpiration pull.**




## Importance of Transpiration

- Absorption of water and minerals from the roots.
- Upward movement of water and minerals from roots to leaves.
- Temperature regulation in plants.

# Transport via Phloem - Translocation

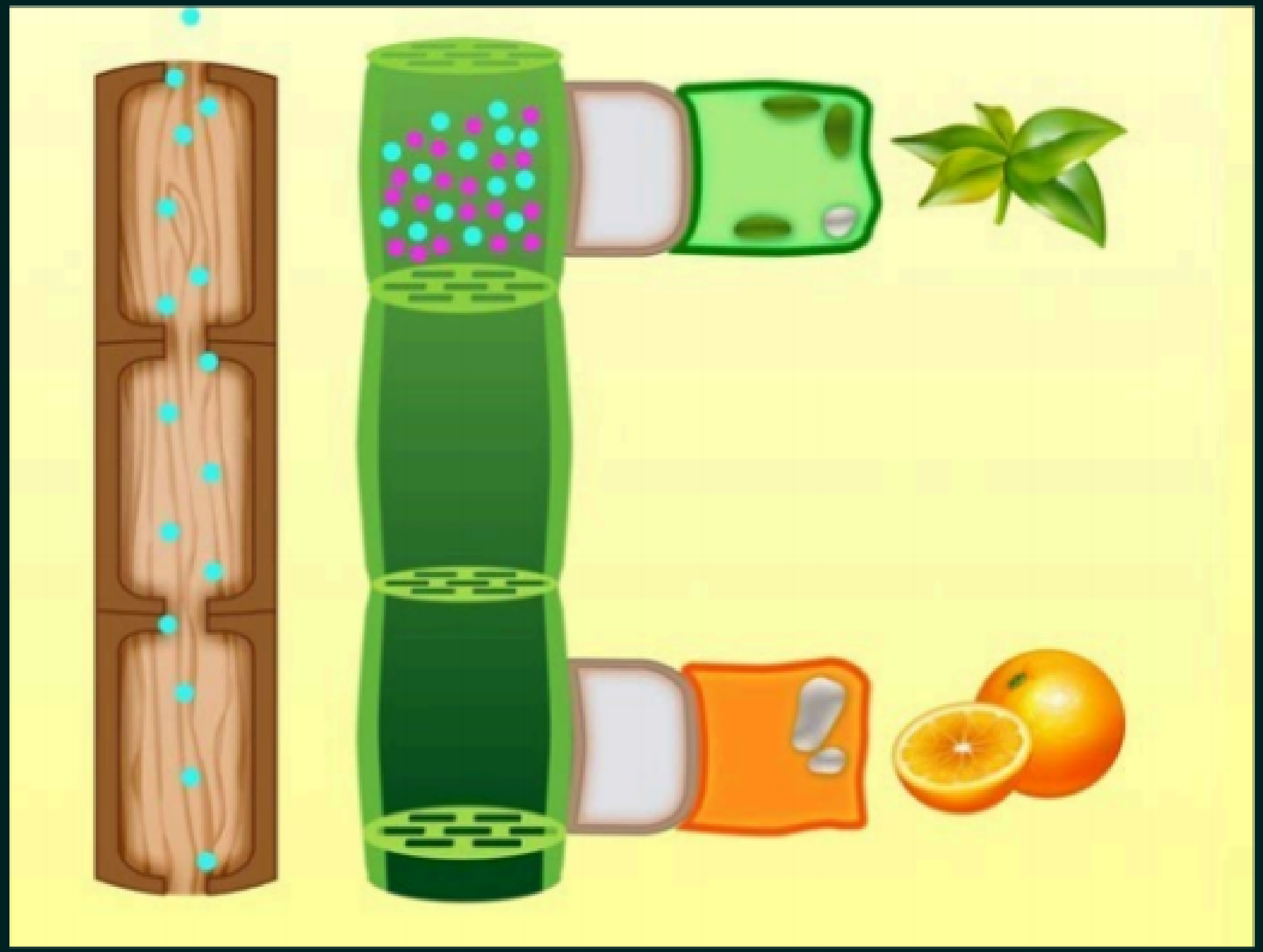
Food made in the leaves is transported to other parts of the plant through the phloem by the process of translocation, which involves active loading of sugars, creation of pressure, and movement of food solution to where it is needed in the plant.

Phloem transports:

- Products of photosynthesis/food
- Amino acids
- Other substances

These substances are transported to:

- Storage organs of roots
- Fruits
- Seeds
- Growing organs



| Feature | Xylem | Phloem |
|---|---|---|
| Function | Transports water and minerals | Transports food (prepared in leaves) |
| Direction of Flow | Unidirectional (only upward) | Bidirectional (both upward and downward) |
| Components | Tracheids, vessels, xylem parenchyma, fibers | Sieve tubes, companion cells, phloem parenchyma, fibers |
| Living/Dead | Mostly dead cells (except xylem parenchyma) | Mostly living cells (except phloem fibers) |
| Role | Helps in water conduction and mechanical support | Helps in transport of food |

# LET'S TEST YOUR MEMORY

Q. Identify the two components of phloem tissue that help in transportation of food in plants.

A. Phloem parenchyma and sieve tubes

B. Sieve tubes and companion cells

C. Phloem parenchyma and companion cells

D. Phloem fibres and phloem parenchyma

# LET'S TEST YOUR MEMORY

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Q. A tall tree continues to transport water even when root pressure is negligible. The most valid explanation is:

80-90

A. Capillary action alone

B. Transpiration pull creates continuous suction

C. Diffusion in xylem

D. Active transport in leaves

# LET'S TEST YOUR MEMORY

Q. Which condition will most strongly disrupt upward water movement?

→ Khatam

A. Increased sunlight

B. Strong wind

C. High humidity + low transpiration

D. High temperature

# LET'S TEST YOUR MEMORY

Q. If phloem transport is blocked, which immediate effect is observed?

A. Water transport stops

B. Food transport to growing parts stops

C. Mineral absorption stops

D. Transpiration increases

# LET'S TEST YOUR MEMORY

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Q. Which statement is most precise?

A. Xylem transport requires energy

B. Phloem transport is passive

C. Xylem is mainly driven by transpiration pull

D. Both use diffusion only

# LET'S TEST YOUR MEMORY

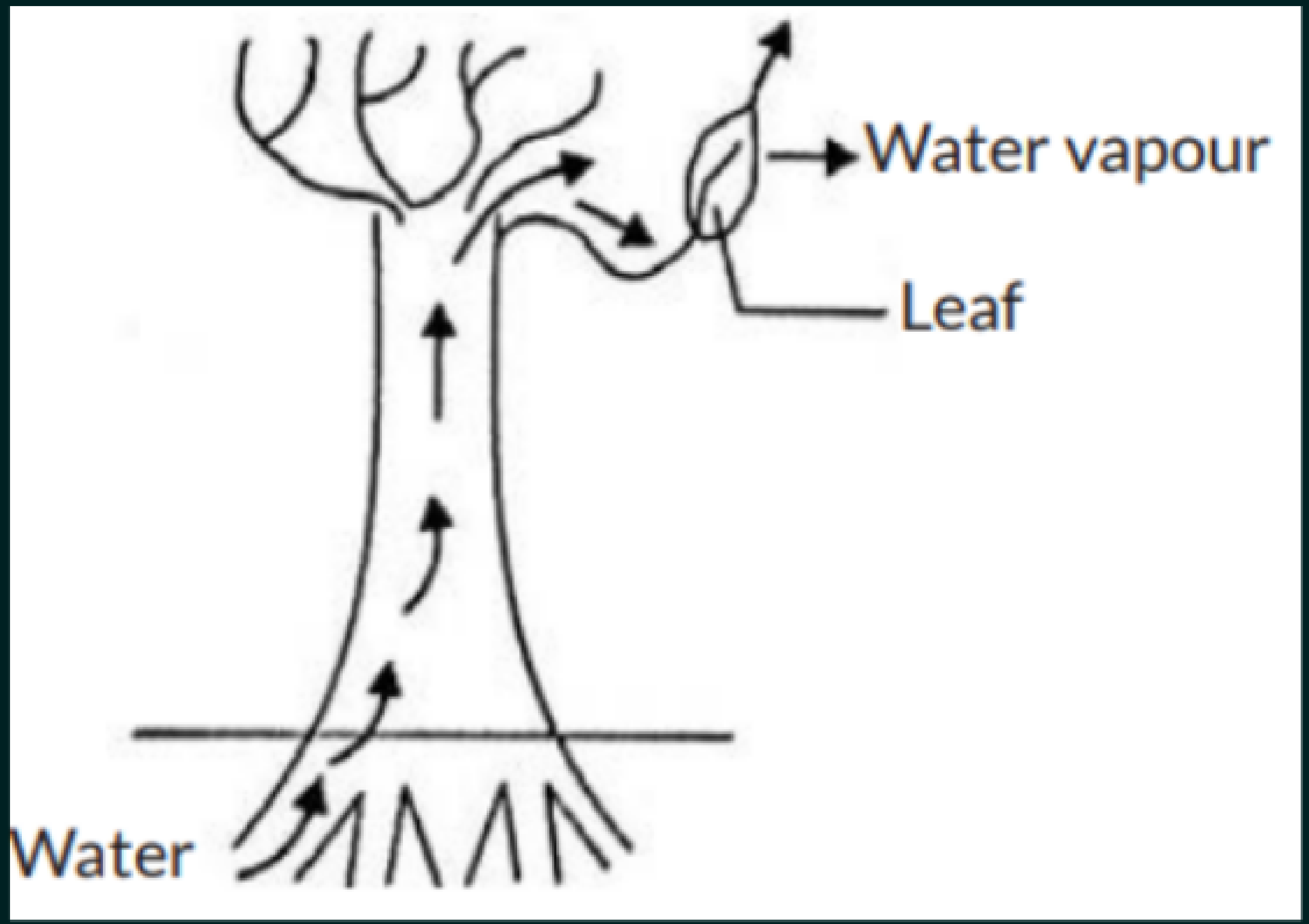
Q. Observe the following diagram and identify the process shown and its significance from the options given below:

(a) Evaporation: Maintains water content in leaf cells.

(b) Transpiration: Creates a suction force which pulls water inside the plant.

(c) Excretion: Helps in excreting waste water from the plant.

(d) Translocation: Helps in transporting materials from one cell to another.



# LET'S TEST YOUR MEMORY

**Q. Why can transpiration be called a "necessary evil"?**

Transpiration is called a "necessary evil" because it causes loss of water from the plant, which can be harmful if water is limited. But it is also necessary because it helps in:

1. Absorption and upward movement of water and minerals from roots to leaves.
2. Cooling the plant in hot weather.
3. Maintaining a continuous transpiration pull for water transport.

So, even though transpiration leads to water loss, it is important for the plant's survival.

# LET'S TEST YOUR MEMORY

Q. Water is used by the leaves of plants for photosynthesis. However, instead of watering the leaves, we water the plant through the soil.

(a) Explain how water absorbed from the soil reaches the leaves of the plant.

(b) Explain the transportation of water and food in plants. How do xylem and phloem function? Also, differentiate between xylem and phloem (any two points).

(c) What is transpiration? Explain its importance. How does transpiration help in the ascent of sap?

(a) Water is absorbed by the roots from the soil and moves upward through xylem tubes in the stem to reach the leaves.

(b) In plants, xylem transports water and minerals from roots to leaves, while phloem transports food prepared in leaves to all parts of the plant. Xylem carries water and minerals, but phloem carries food. Xylem transport is mainly upward, while phloem transport can be upward and downward.

(c) Transpiration is the loss of water as water vapour from leaves. It helps in pulling water upward from roots to leaves and also cools the plant. This upward movement of water is called ascent of sap.

# LET'S TEST YOUR MEMORY

| Xylem | Phloem |
|---|---|
| Transports water and minerals | Transports food |
| Transport is upward only | Transport is bidirectional |
| Made mainly of dead cells | Made mainly of living cells |

(c) Transpiration is the loss of water in the form of water vapour from aerial parts of the plant through stomata.

Importance of Transpiration:

1. It creates transpiration pull, helping in ascent of sap.
2. It cools the plant.
3. It helps in absorption and transport of minerals.

Role in Ascent of Sap: When water evaporates from leaves, a suction force is created. This force pulls water upward through xylem from roots to leaves.


Bye!